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New Insights on the Geometry of Simultaneous Multiple Fracture Propagation in Laminated Shale Reservoirs Considering the Effects of Bedding Planes and Stress Interference

Rui He, Yu Sang, Li Li, Weihua Chen, Ji Zeng, Tao Wang, and Feng Zhao, Engineering Technology Research Institute, PetroChina Southwest Oil & Gasfield Company, Chengdu, Sichuan, China / Sichuan Key Laboratory of Shale Gas Geological Evaluation and Efficient Development, Chengdu, Sichuan, China; Liuke Huang, School of Civil Engineering and Geomatics, Southwest Petroleum University, Chengdu, Sichuan, China

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Abstract

Multi-stage horizontal well fracturing has been widely applied to exploit unconventional shale oil & gas resources. The shale reservoir generally contains a large number of bedding planes with lower strength, which have a great effect on the growth of hydraulic fractures. In this paper, the 3D discrete lattice method is applied to establish a 3D model of simultaneous multiple fracture propagation for laminated shale reservoirs. This model considers the influence of stress interference, perforation friction, fluid-mechanical coupling, and bedding planes. The effects of geological and engineering parameters on multiple fracture propagation are studied. The bedding plane activation efficiency and variation coefficient of fracture length are both proposed to quantitatively evaluate the stimulating effect. The simulation results show that the geometry of each fracture within a stage is obviously different. The existence of bedding planes will increase the fluid filtration, resulting in reduction of fracture geometry. Under the combined effect of bedding planes and stress interference, the bedding planes are more easily activated by hydraulic fracture under the condition of low bedding cohesion, low vertical-horizontal stress difference, small cluster space, high injection rate, and low fracturing fluid viscosity. However, those factors have a smaller impact on the fracture uniformity. The research results of this paper provide an effective method for accurately predicting the geometry of simultaneously propagated fractures in laminated shale reservoirs for horizontal wells. This method is applied in Well L1 in the Eastern Sichuan Basin, China, and the monitoring results indicate that a complex fracture system has been formed through hydraulic fracturing operation, and ideal productivity has been achieved.

Introduction

As an unconventional resource, shale oil has changed the world's energy structure. China, possessing vast shale oil resources, has recently made significant breakthroughs in shale oil exploration and development within multiple basins, including the Ordos, Junggar, Bohai Bay, Santanghu, Sichuan, and Qaidam Basins

(Fu et al., 2022). Differing with marine shale oil reservoirs in the United States, China's lacustrine terrigenous organic-rich shales exhibit significantly more diverse hydrocarbon enrichment patterns, widely distributed across both eastern and western continental basins (Jiao et al., 2020). Chinese shale oil reservoirs are characterized by low porosity and ultra-low permeability, making multi-stage clustered fracturing in horizontal wells an essential technique for their development (Lei et al., 2022). Abundant bedding planes are observed in some shale formations, typically exhibiting horizontal orientation with interlayer spacing ranging from centimeter to meter scale. As mechanically weak discontinuities, the bedding planes provide channels for crude oil flow, however, the mechanical strength of the bedding plane is significantly lower than that of the reservoir matrix. When encountering bedding planes during the propagation of hydraulic fracture, the bedding planes will limit the vertical growth of hydraulic fracture, resulting in insufficient fracture height and affecting the effectiveness of reservoir stimulation (He et al., 2023). In recent years, the influence of bedding planes on hydraulic fracture propagation has become a research focus.

Several researchers have investigated the effects of discontinuities on fracture propagation through laboratory experiments, particularly using true triaxial fracturing tests. This method can intuitively observe the fracture morphology and obtain some understanding of the mechanism. Employing combined cubic samples, Teufel and Clark (1981) investigated the control of interfacial shear strength and elastic modulus difference on fracture height growth. Heng et al. (2015) demonstrated that bedding planes' low fracture toughness from weak cementation preferentially inhibits fracture propagation. Their results revealed that fracture network development primarily occurs through hydraulic fractures branching and reorienting along bedding planes. Hou et al. (2018) performed true triaxial experiments with acoustic emission (AE) monitoring to analyze fracture initiation and propagation in deep shales. The results show that the bedding plane has an obvious impact on the evolution of fracture network, and four distinct fracture morphologies were categorized. Wang et al. (2021) conducted comparative physical experiments using downhole full-diameter cores from the Bohai Bay Basin to investigate fracture propagation patterns in sandstone and shales. Their results revealed three typical fracture morphologies: planar I-shaped fractures in naturally fractured sandstone, cross-shaped fractures in tight sandstone without natural fractures, and stepped fracture networks in laminated shale formations.

While laboratory experiments are not only costly but also limited in scale, numerical simulation has emerged as a crucial alternative approach for investigating fracture propagation. Conventional two-dimensional (2D) models, including the PKN model (Nordgren et al., 1972), and KGD model (Geertsma et al., 1969), along with subsequently proposed pseudo-three dimensional (P3D) models (Dontsov et al., 2022), fail to adequately account for fracture height. These models consequently cannot satisfy the requirements for simulating fracture propagation while considering the effects of bedding planes. With advancements in computational technology, novel numerical simulation methods have emerged as powerful tools for investigating fracture propagation under the influence of bedding planes. Zheng et al. (2025) developed a 3D hydraulic fracturing model based on the continuous-discrete element method (CDEM) to investigate fracture-bedding plane interaction. Their simulations revealed that net pressure reduction primarily controls vertical fracture propagation. Through the cohesive zone model (CZM), Li et al. (2025) studied the interaction between a hydraulic fracture and bedding planes with consideration of softening effects and plastic zones. Their analysis further examined the induced stress distributions generated by both hydraulic fractures and bedding fractures. Li et al. (2025) integrated modeling approach combined a BP sliding zone lumped model with planar 3D fracturing simulation, identifying interlayer stress difference as the primary control on vertical fracture propagation and showing BP slip extent to be affected by both stress anisotropy and friction coefficient. Chen et al. (2025) applied finite element method (FEM) incorporating bedding planes to analyze hydraulic fracture height growth, and they systematically evaluated how fracture propagation behavior is influenced by bedding dip angle, interfacial strength, stress anisotropy, and Young's modulus difference between reservoir and barrier layers.

As a discontinuous media modeling technique, discrete element method (DEM) is ideally suited for large deformation analysis in naturally fractured shale reservoirs (Detournay et al., 2022). Through 3D block DEM method, He et al. identified four key penetration-enhancing factors: high vertical stress, strong bedding cohesion, viscous fluids, and large injection rate. The hydraulic fracture-bedding plane interaction typically includes "キ", "エ", and "I" shapes, reflecting complex fracture geometry in shales (He et al., 2023). The 3DEC-based investigation by Zheng et al. (2022) developed a dimensionless parameter T to evaluate the penetration ability under different engineering parameters for laminated shales. The above research mainly focuses on exploring the influence of one or more bedding planes on the propagation of a single fracture from a mechanistic perspective. However, in fact, multiple fractures propagate simultaneously within each stage in horizontal well fracturing, and there is a lack of research on the multiple fracture propagation considering the bedding planes.

In this paper, we used the 3D discrete lattice method to establish a numerical model of simultaneously propagated fractures for laminated shale reservoirs. This model considers the influence of stress interference, perforation friction, fluid-mechanical coupling, and bedding planes. Based on this model, the fracture geometries under different geological and engineering parameters are analyzed. The bedding plane activation efficiency and the variation coefficient of fracture length are both proposed to quantitatively evaluate the stimulating effect. The research results allow us to gain a better insight to predict the multiple fracture growth of hydraulic fracture in naturally-layered shale reservoirs, which is beneficial to the design of hydraulic fracturing parameters in shale oil reservoirs.

Methodology

3D discrete lattice method

As a kind of DEM, the 3D discrete lattice method is a simplified bonded particle model (BPM) (Potyondy et al., 2004) based on discrete lattice theory and synthetic rock mass technology, which can effectively describe the propagation of hydraulic fractures (Benouadah et al., 2021). The synthetic rock mass technology employs the BPM to simulate rock deformation and utilizes a smooth joint model (SJM) to characterize rock mass joints. Compared to conventional DEM, the 3D discrete lattice method simplifies the BPM through a lattice network. It replaces particles in the BPM with mass-bearing nodes and connects adjacent nodes via springs to simulate contacts (analogous to cementation between particles). This approach allows natural joints to be imported into the lattice network at arbitrary angles while significantly increasing mesh density and improving computational efficiency (Zhang et al., 2021), as shown in Fig. 1. The algorithm incorporates techniques such as embedded lattices and smooth joint elements, enabling effective simulation of nonlinear behaviors in joints, including rupture, slip, opening, and closure. It demonstrates distinct advantages in handling interactions between natural joints and hydraulic fractures.

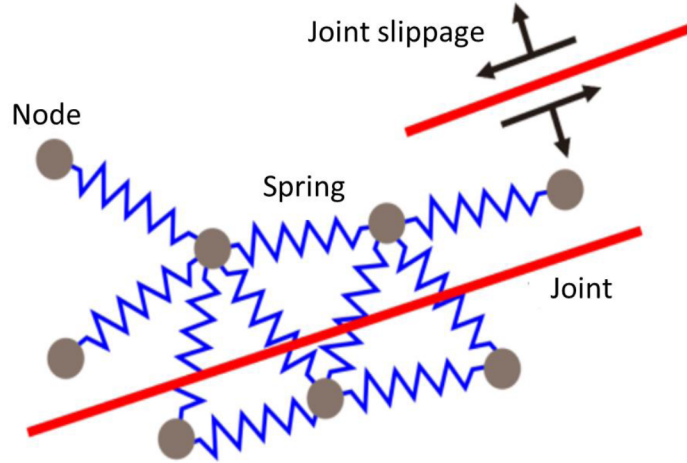


Figure 1—Schematic diagram of grid network (Cundall et al., 2011).

In the 3D discrete lattice method, each node's motion is determined by both its spatial position and contact forces. Governed by Newton's second law, the translational and rotational equations of motion for each node are defined as:

$$\begin{cases} F = ma \\ M = I\omega \end{cases} \quad (1)$$

where F is the resultant force acting on the node, N, m is the node mass, kg, a is the node acceleration, m/s², M is the resultant moment acting on the node, N·m, I is the inertia moment of nodes, kg·m², and ω is the node angular velocity, rad/s.

The acceleration of a node at time t is computed explicitly through the intermediate velocity at $t + \Delta t$, governed by:

$$a^t = \frac{v^{t+\Delta t/2} - v^{t-\Delta t/2}}{\Delta t} \quad (2)$$

where a^t is the acceleration of a node at time t , m²/s, and Δt is the time step, s.

The degrees of freedom at each node can be iteratively solved using the central difference formula, with the translational degree of freedom to update equations, as follows:

$$\begin{cases} v_i^{t+\Delta t/2} = v_i^{t-\Delta t/2} + \sum F_i^t \Delta t / m \\ u_i^{t+\Delta t/2} = u_i^{t-\Delta t/2} + v_i^{t+\Delta t/2} \Delta t \end{cases} \quad (3)$$

where v_i^t is the velocity of component i ($i=1,2,3$) at time t , m/s, F_i^t is the displacement of component i at time t , m, u_i^t is the force of component i acting on the node at time t , N.

The rotational degrees of freedom are updated by the following equation:

$$\begin{cases} \omega_i^{t+\Delta t/2} = \omega_i^{t-\Delta t/2} + \sum M_i^t \Delta t / I \\ I = 2mR^2 / 5 \end{cases} \quad (4)$$

where ω_i^t is the angular velocity of component i at time t , rad/s, M_i^t is the moment of component i at time t , N·m, and R is the equivalent radius of nodes, m.

Based on node relative displacement, the normal and tangential forces of the spring can be updated as:

$$\begin{cases} \Delta F_i^N = v_i^N k^N \Delta t \\ \Delta F_i^S = v_i^S k^S \Delta t \end{cases} \quad (5)$$

where ΔF_i^N is the increased normal force of component i, N, ΔF_i^S is the increased tangential force of component i, N, v_i^N is the normal velocity of component i, m/s, v_i^S is the tangential velocity of component i, m/s, k^N is the normal stiffness of spring, N/m, and k^S is the tangential stiffness of spring, N/m.

The correspondence between the tensile strength and shear strength of microscopic spring and those of macroscopic rock masses are as follows:

$$\begin{cases} F^{N,\max} = \alpha_t T R^2 \\ F^{S,\max} = \mu F^{N,\max} + \alpha_s C R^2 \end{cases} \quad (6)$$

where $F^{N,\max}$ and $F^{S,\max}$ are the tension force and shear force of the spring, respectively, N, α_t and α_s are the correction coefficients for tensile and shear strength, respectively, T and C are the tensile and shear strength of macroscopic rock mass, respectively, Pa, R is the grid cell size, m, and μ is the friction coefficient.

The spring element undergoes tensile failure when its normal stress exceeds the tensile strength, or shear failure when the tangential force surpasses the shear strength. Upon spring failure, micro-fractures initiate, and both the normal and tangential forces of the failed spring element are reset to zero.

$$q = \beta k_r \frac{W^3}{12\mu} [p_A - p_B + \rho_w g (z_A - z_B)] \quad (7)$$

where q is the flow rate, m³/s, β is the dimensionless correction factor, k_r is the relative permeability, W is the fracture width, m, p_A , p_B are the fluid pressure at node A and node B, respectively, Pa, ρ_w is the fluid density, kg/m³, g is the gravitational acceleration, m/s², and z_A , z_B are the height of node A and node B, respectively, m.

The relative permeability is a function of saturation, and when the pipeline is saturated, the relative permeability is 1, and the relationship between them can be expressed as:

$$K_r = S^2(3 - 2S) \quad (8)$$

where K_r is the relative permeability, and S is the fluid saturation.

In the time step of flow, the fluid pressure increment is expressed as:

$$\Delta p = \frac{Q}{V} K_f \Delta t_f \quad (9)$$

where Δp is the increment of fluid pressure, Pa, K_f is the bulk modulus of fluid, GPa, V is the fluid volume at nodes, m³, Q is the sum of all flows connected to the node pipeline, m³/s, and Δt_f is the time step of fluid flow, s.

Q can be expressed as:

$$Q = \sum_{i=1}^n q_i \quad (10)$$

where q_i is flow rate of pipeline connected to node i, m³/s.

The q_i is calculated according to Darcy's law:

$$q_i = \frac{(p_i^{\text{in}} - p_i^{\text{out}}) s^2}{k_f \Delta t} \quad (11)$$

where p_i^{in} is the pressure at the inlet of the fluid unit, Pa, p_i^{out} is the pressure at the outlet of the fluid unit, Pa, s is the area of fluid unit, m², and k_f is the apparent mechanical stiffness of fluid units, N/m.

The proposed model achieves complete two-way coupling between mechanical and fluid systems through simultaneous resolution of fluid transport, fracture initiation/propagation, and rock mass deformation. The coupling mechanism operates through three primary interactions: (1) Fluid pressure acts on mechanical model nodes and fluid elements, inducing rock deformation and fracture, leading to tensile opening or shear

sliding of cracks. (2) Rock deformation and failure change the crack opening and dynamically regulate the permeability field of the crack network. (3) The fluid pressure field updates in real-time with the rock deformation, satisfying the strong coupling condition of the mass momentum conservation equation. When the stress intensity factor at the crack tip is significantly lower than the fracture toughness, the spring tensile strength threshold is used to determine crack propagation. When the stress intensity factor at the tip approaches the fracture toughness, the crack propagation path is directly determined based on the fracture toughness criterion.

Model verification

The analytical solution under the same conditions was applied to verify the accuracy of the proposed 3D discrete lattice method. Both approaches simulate the propagation of a penny-shaped hydraulic fracture under viscosity-dominated condition, without considering fluid leak-off. Fig. 2 shows the comparison between the theoretical solution and the numerical results for penny-shaped fracture. The results indicate that the numerical and theoretical solutions are basically consistent, verifying the effectiveness of the 3D discrete lattice method.

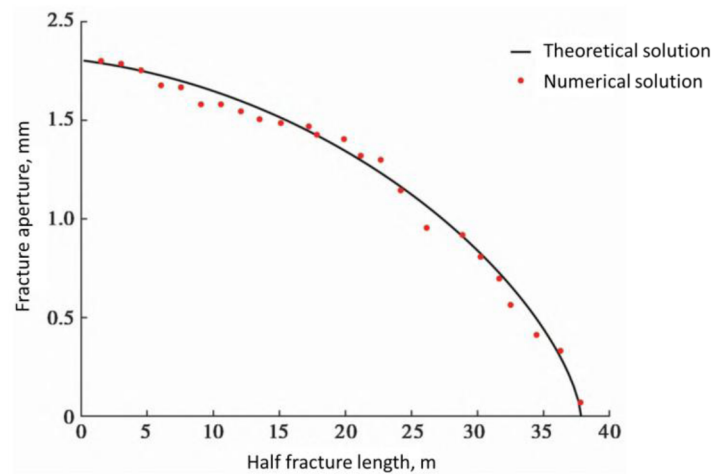


Figure 2—Comparison between the theoretical solution and the numerical results for penny-shaped fracture (Huang et al., 2024).

Using the 3D discrete lattice method, Fu et al. (2022) established a shale model with two horizontal bedding planes at the same scale as the laboratory true triaxial experiment. The model incorporates a horizontal wellbore flanked by two rectangular bedding planes (dimensions: 600mm×800mm; vertical spacing: 150mm from the wellbore). The mechanical strength of the bedding plane is lower than that of the matrix. The geo-stress field was set to strike slip mode, and the low-viscosity slick-water (5mPa·s) was applied. Both experimental and numerical results demonstrate consistent fracture patterns (Fig. 3), where hydraulic fractures partially penetrate the upper and lower bedding planes. Unlike the penny-shaped fractures observed in homogeneous models, the presence of bedding planes leads to elliptical fracture propagation aligned with the maximum principal stress direction. These findings validate the 3D discrete lattice method for simulating fracture propagation in laminated shale reservoirs.

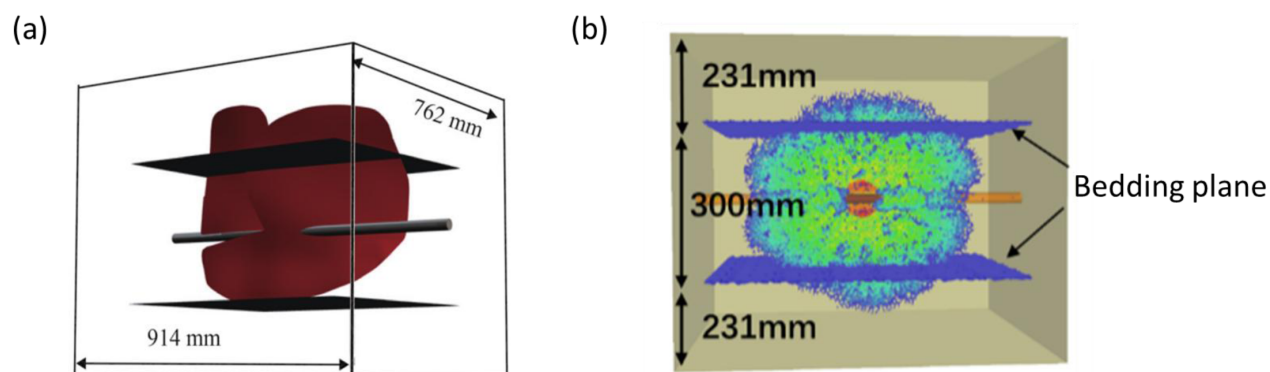


Figure 3—Model validation by true triaxial experiment for shale with two horizontal bedding planes: (a) Experimental results obtained by laser scanning; (b) Numerical results by 3D discrete lattice method (Fu et al., 2022).

Modeling scheme

Model for horizontal well fracturing

To systematically investigate the multiple fracture propagation laws in horizontal wells under the combined influence of bedding planes and stress interference, this study takes the Jurassic Lianggaoshan Formation laminated shale reservoir in the Sichuan Basin as an example. A horizontal well fracturing model was established based on the 3D discrete lattice method, with model dimensions of $90\text{m} \times 30\text{m} \times 30\text{m}$, as shown in Fig. 4. Five perforation clusters are set on the horizontal well, and hydraulic fractures propagate from these perforation clusters. Three horizontal bedding planes are pre-set above and below the wellbore. Using this model, the study examines fracture propagation patterns under various geological parameters (vertical-horizontal stress difference and bedding cohesion) and engineering parameters (cluster space, injection rate, and fracturing fluid viscosity). The input parameters for numerical simulations are listed in Table 1. For each simulation, the baseline input parameters remain constant. Specifically, the Young's modulus is 45GPa , Poisson's ratio is 0.22 , the reservoir thickness is 30m , the fracture toughness is $1\text{MPa}\cdot\text{m}^{0.5}$, the number of perforations hole is 40 , and the internal friction angle of bedding plane is 20° .

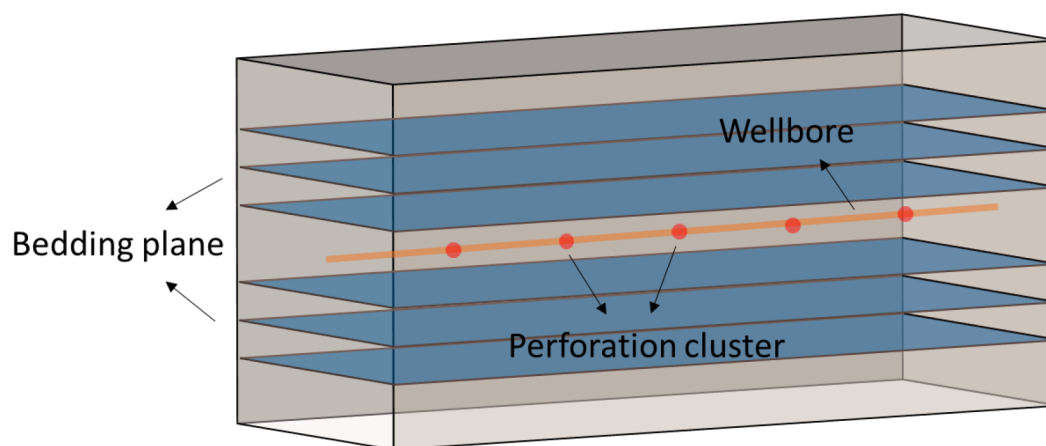


Figure 4—The established model of shale oil reservoir with horizontal bedding planes.

Table 1—The basic input parameters for multiple fracture propagation simulation of laminated shale.

Parameters	Value	Unit
Young's modulus	45	GPa
Poisson's ratio	0.22	/
Reservoir thickness	30	m
Fracture toughness	1	MPa·m ^{0.5}
Cluster number per stage	5	/
Perforation hole number per stage	40	/
Internal friction angle of bedding plane	20	°
Vertical-horizontal stress difference	2/4/6/8	MPa
Bedding cohesion	0.5/2/3.5/5	MPa
Cluster space	6/9/12/15	m
Injection rate	6/10/14/18	m ³ /min
Fracturing fluid viscosity	3/5/30/100	mPa·s

Evaluation method

Hydraulic fracturing aims to maximize the stimulated reservoir volume (SRV). For horizontal wells in laminated shales, the fence-like fracture network containing hydraulic fractures and activated bedding planes will be generated. In addition to focusing on the penetration of hydraulic fracture, special attention should be paid to fracture length and the activation of bedding planes.

For multi-cluster staged fracturing, engineers primarily aim to achieve uniform fracture lengths among clusters to ensure a larger oil drainage area. In this study, the variation coefficient of fracture length is employed to quantitatively evaluate the fracture uniformity (see Equation 12). A lower variation coefficient of fracture length indicates more balanced fracture growth and better fracturing performance.

$$\eta = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (L_i - \bar{L}_i)^2}}{\bar{L}_i} \times 100\% \quad (12)$$

where η is the variation coefficient of fracture length, N is the fracture number within a stage, L_i is the fracture length of the i -th perforation cluster, m, and $\bar{L}_i$ is the average fracture length within a stage, m.

Quantitative characterization of fracture complexity is a crucial issue in hydraulic fracturing. Generally, higher fracture complexity leads to a larger SRV and better fracturing performance. This study proposes a fracture area-based method to quantify fracture complexity, using bedding plane activation efficiency as the indicator to evaluate fracture complexity. The equation is as follows:

$$\beta = \frac{\sum_{i=1}^n A_i}{A_t} \times 100\% \quad (13)$$

where β is the bedding plane activation efficiency, %, A_i is the activated area of the i -th bedding plane, m², n is the number of pre-setting bedding plane, and A_t is the total areal of bedding planes, m².

Modeling results and discussion

Basic case analysis

This section presents a comparative analysis of multi-fracture propagation patterns under two scenarios: with and without bedding planes, as illustrated in Fig. 5. Fig. 5a displays simulated fractures in a homogeneous reservoir (without bedding planes), where five approximately parallel hydraulic fractures perpendicular to the wellbore are formed, propagating along the maximum principal stress direction. Due to stress interference between adjacent fractures, the inner fractures are inhibited by the outer fractures to some extent, resulting in relatively reduced length, width, and height of the inner fractures compared to the outer fractures. Fig. 5b shows simulated fractures in a laminated reservoir (with bedding planes), demonstrating the generation of a fence-like complex fracture network consisting of five main fractures and partially activated horizontal bedding planes. Compared with the simulation results of homogeneous reservoirs (Fig. 5a), the fracture height and length are both confined at different levels due to the fluid filtration into bedding planes. The outer fractures exhibit an obvious inhibitory effect on the inner two fractures, demonstrating that the existence of bedding planes enhances stress interference effects. Meanwhile, the activated bedding planes display notably limited widths. This phenomenon occurs because when a hydraulic fracture approaches a bedding plane, the normal stress decreases while the shear stress increases near their intersection line (Zheng et al., 2022). Under these conditions, as fracturing fluid flows from the hydraulic fracture into the bedding planes, shear failure becomes more likely to occur along the bedding plane. Consequently, while the bedding planes become activated and tend to displace along the shear direction, they do not experience significant opening. Therefore, we recommend employing proppants with smaller particle sizes to effectively support these opened bedding planes during fracturing operation.

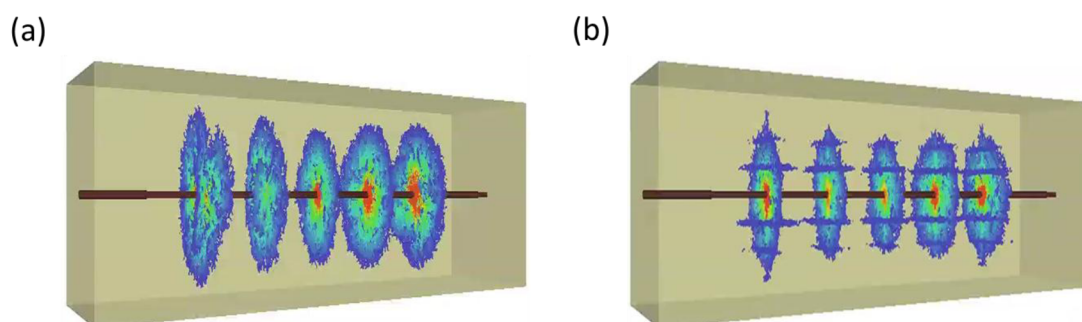


Figure 5—Simulation results of multiple fracture propagation for 5 clusters: (a) Without bedding planes; and (b) With bedding planes.

Effect of geological parameters

Vertical-horizontal stress difference. In this section, the geometries of multiple fracture propagation under various vertical-horizontal stress difference (the difference between vertical stress and minimum horizontal principal stress) in laminated shale is studied, as shown in Fig. 6, and quantitative evaluation results are shown in Fig. 7. The results demonstrate that under varying vertical-horizontal stress difference, the inner fractures are inhibited by the outer fractures to some extent, resulting in smaller geometries for the inner fractures. As the vertical-horizontal stress difference increases, the activated area of bedding planes decreases. When the vertical –horizontal stress difference rises from 2MPa to 6MPa, the bedding plane activation efficiency declines from 16.3% to 10.4%. This is attributed to the enhanced clamping effect of stress on the bedding planes under higher vertical stress, making shear failure and fracture deflection more difficult to occur, thereby increasing average fracture height from 20.6m to 22.5m. When the vertical-horizontal stress difference exceeds 6MPa, its influence on the activated bedding area and fracture height becomes very small. On the other hand, as the vertical-horizontal stress difference increases, the variation

coefficient of fracture length increases from 8.7% to 15.4%. However, when the vertical-horizontal stress difference exceeds 6 MPa, its impact on the fracture uniformity is less.

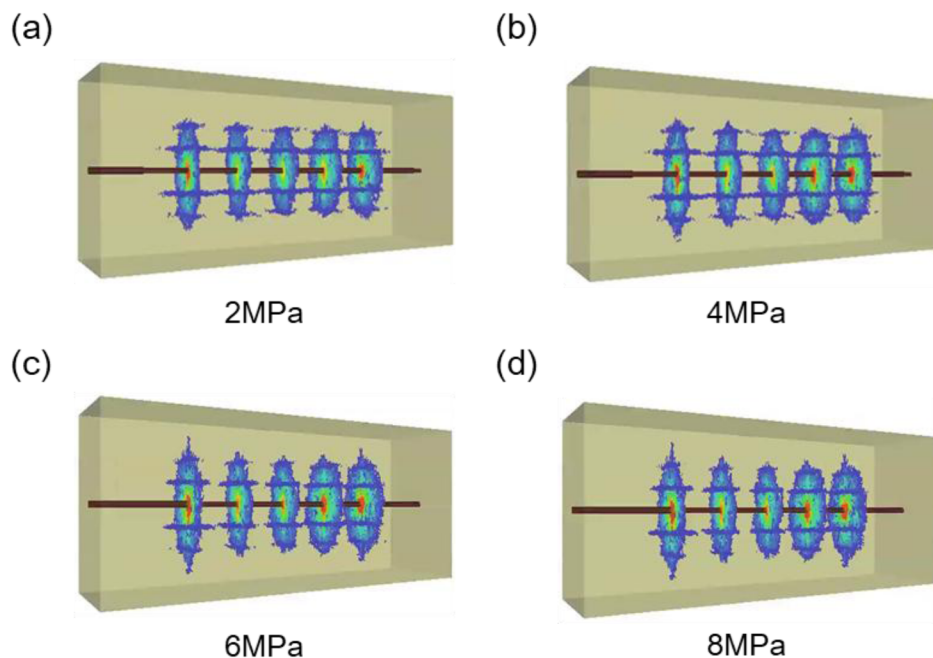


Figure 6—The simulated fracture geometry under various vertical-horizontal stress difference: (a) 2MPa; (b) 4MPa; (c) 6MPa; and (d) 8MPa.

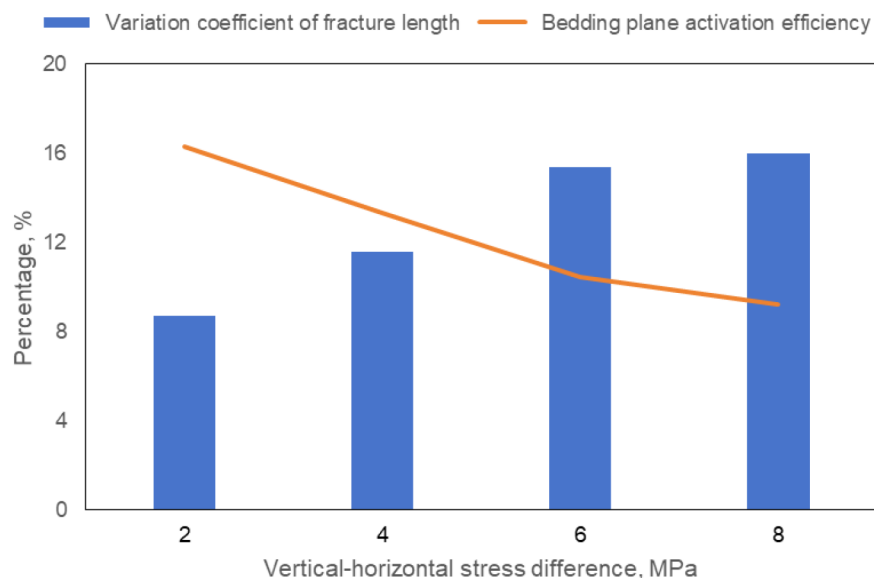


Figure 7—Quantitative evaluation of multiple fracture propagation under various vertical-horizontal stress difference: (a) 2MPa; (b) 4MPa; (c) 6MPa; and (d) 8MPa.

Bedding cohesion. In geotechnical mechanics, cohesion reflects the inherent bonding force between particles, and bedding cohesion represents the shear resistance of bedding planes. In this section, the geometries of multiple fracture propagation under various bedding cohesion in laminated shale is studied, as shown in Fig. 8, and quantitative evaluation results are shown in Fig. 9. With increasing bedding cohesion, the activated bedding area progressively decreases due to the greater difficulty for shear failure to occur at bedding planes. As the bedding cohesion rises from 0.5 to 5MPa, the bedding plane activation efficiency declines from 12.6% to 7.2%. Meanwhile, the ability of vertical fracture propagation is enhanced, and the

fracture height increases from 21.8m to 23.7m. On the other hand, the variation coefficient of fracture length initially increases and subsequently decreases with the increase of bedding cohesion, but the overall fracture uniformity remains relatively stable, with the variation coefficient of fracture length maintaining within the range of 14-16%.

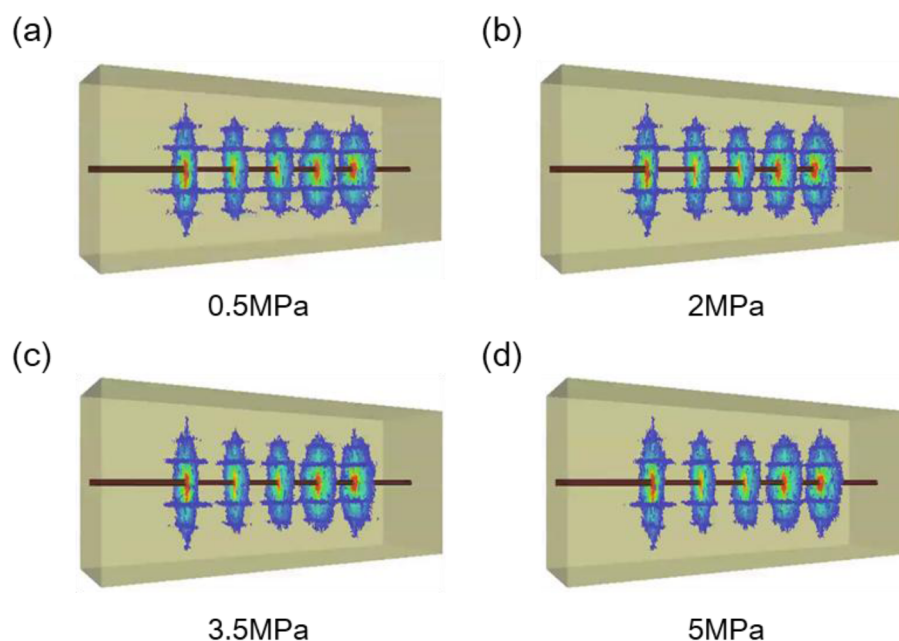


Figure 8—The simulated fracture geometry under various bedding cohesion: (a) 0.5MPa; (b) 2MPa; (c) 3.5MPa; and (d) 5MPa.

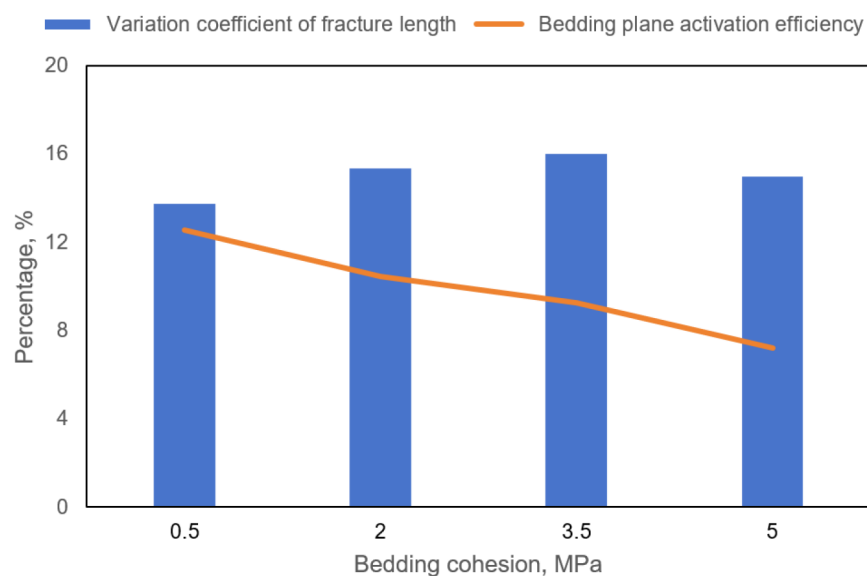


Figure 9—Quantitative evaluation of multiple fracture propagation under various bedding cohesion: (a) 0.5MPa; (b) 2MPa; (c) 3.5MPa; and (d) 5MPa.

Effect of engineering parameters

Cluster space. Cluster space is one of the important engineering factors affecting the multiple fracture propagation in horizontal well fracturing. Appropriate cluster space design can effectively improve the fracture geometries and obtain good stimulating results. In this section, the geometries of multiple fracture propagation under various cluster space in laminated shale is studied, as shown in Fig. 10, and quantitative evaluation results are shown in Fig. 11. As the cluster space decreases, the activated bedding area increases

slightly. When the cluster space is reduced from 15 m to 6 m, the bedding plane activation efficiency increases from 9.1% to 12.8%, while the average fracture height decreases from 23.3m to 20.8m. This suggests under the combined effects of stress interference and bedding planes, the shear failure is easier to occur at bedding planes, which exerts a certain confining effect on fracture height growth and promotes fracture deflection along bedding planes. On the other hand, when the cluster space is reduced from 15m to 9m, the variation coefficient of fracture length remains nearly unchanged. However, as the cluster space further decreases from 9m to 6m, this coefficient sharply increases from 15.4% to 27.1%. This indicates that the existence of bedding planes may change the stress field at lower cluster space, resulting in more significant non-uniform fracture propagation.

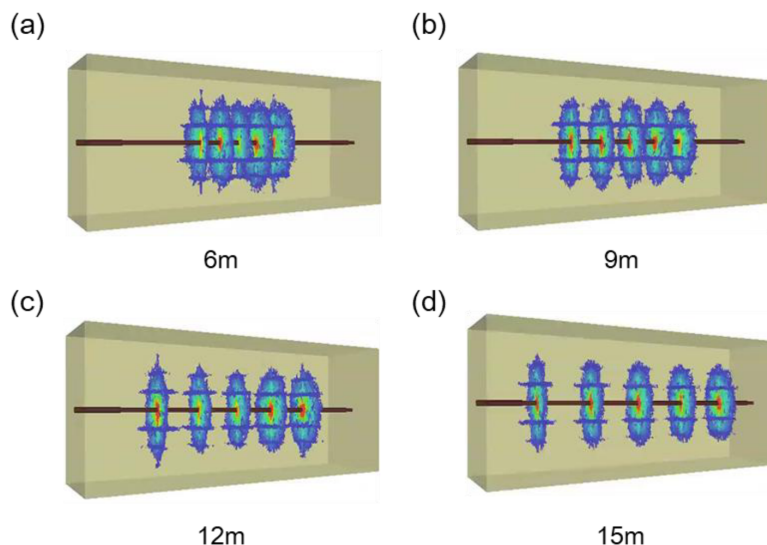


Figure 10—The simulated fracture geometry under various cluster space: (a) 6m; (b) 9m; (c) 12m; and (d) 15m.

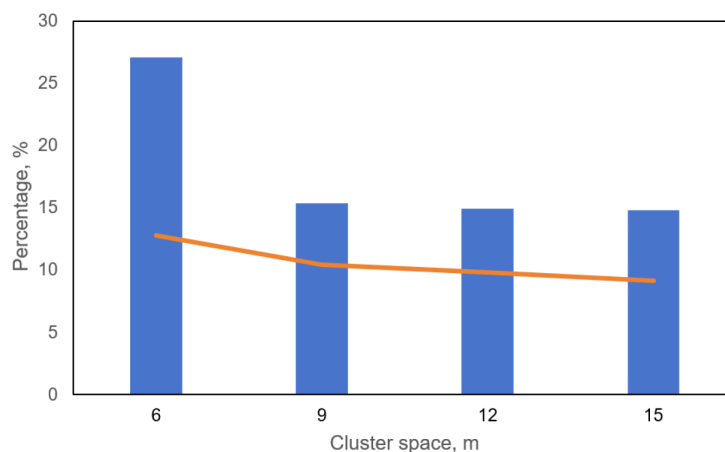


Figure 11—quantitative evaluation of multiple fracture propagation under various cluster space: (a) 6m; (b) 9m; (c) 12m; and (d) 15m.

Injection rate. Injection rate is another important engineering factor affecting fracture propagation and proppant migration. The larger the injection rate, the more favorable it is for fracturing operation. However, it requires higher hydraulic power and costs more, and the impact of injection rate on multiple fracture propagation under the influence of bedding planes is unclear. In this section, the geometries of multiple fracture propagation under various injection rates in laminated shale is studied, as shown in Fig. 12, and quantitative evaluation results are shown in Fig. 13. The simulation results indicate that higher injection rate activates larger area of bedding planes. As the injection rate increases from $6\text{m}^3/\text{min}$ to $18\text{m}^3/\text{min}$, the

bedding plane activation efficiency rises from 8.3% to 11.4%. Analysis suggests that at higher injection rate, the net pressure inside the fracture increases, and the shear force acting on the bedding plane increases. According to the Mohr-Coulomb criterion, the bedding plane is more prone to shear slip, meanwhile at higher injection rate, the fluid is more likely to filter out and enter the bedding planes, promoting their opening. In addition, with the increase of injection rate, the ability of fracture penetration is greatly improved, and the average fracture height increases significantly from 18m to 25m. Analysis indicates that higher net pressure under higher injection rate alters the stress field near the fracture tip. Under this condition, it is easier for hydraulic fractures to penetrate the bedding planes. Nevertheless, excessively high injection rate may result in uncontrolled fracture height growth and reduced bedding plane activation efficiency. Consequently, optimal fracture network complexity requires maintaining injection rates within an intermediate range during stimulation treatment. From the simulation results, injection rate has little effect on the fracture uniformity, and the variation coefficient of fracture length under various injection rate is around 15%.

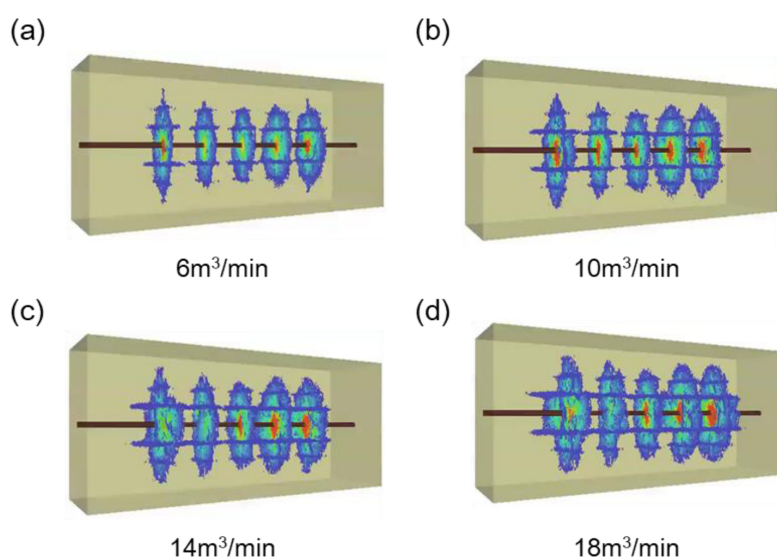


Figure 12—The simulated fracture geometry under various injection rate: (a) 6m³/min; (b) 10m³/min; (c) 14m³/min; and (d) 18m³/min.

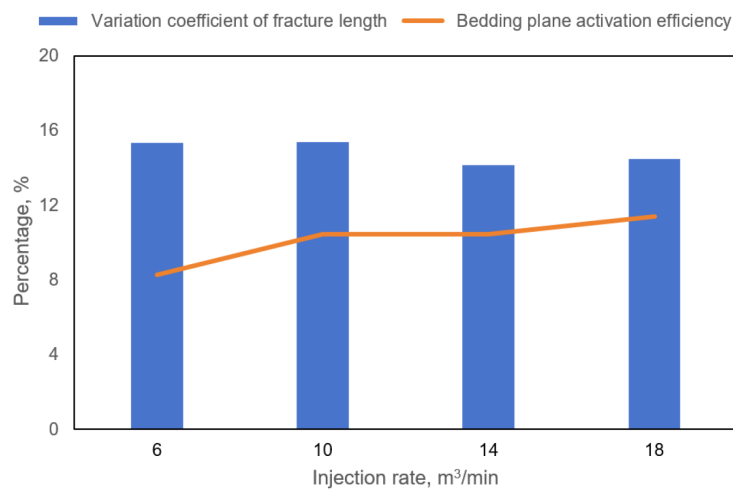


Figure 13—Quantitative evaluation of multiple fracture propagation under various injection rate: (a) 6m³/min; (b) 10m³/min; (c) 14m³/min; and (d) 18m³/min.

Fracturing fluid viscosity. In this section, the geometries of multiple fracture propagation under various fracturing fluid viscosity in laminated shale is studied, as shown in Fig. 14, and quantitative evaluation results are shown in Fig. 15. The simulation demonstrates an inverse correlation between fracturing fluid viscosity and bedding plane activation efficiency. As fracturing fluid viscosity increases from 3mPa·s to 100mPa·s, the bedding plane activation efficiency decreases from 11.6% to 8.4%. This reduction can be attributed to two primary mechanisms under high-viscosity conditions: significantly impaired fluid leak-off into bedding planes, and rigid fracture propagation which is easier to penetrate the bedding planes rather than activating them. Furthermore, elevated viscosity leads to increased fracture width but reduced fracture length under high net-pressure conditions. From the simulation results, fracturing fluid viscosity has little effect on the fracture uniformity, and the variation coefficient of fracture length only increases slightly with the increase of fracturing fluid viscosity.

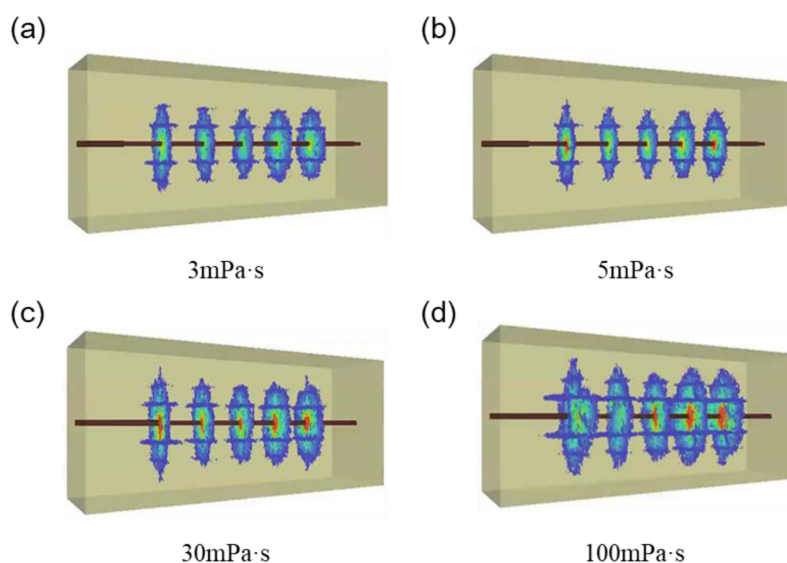


Figure 14—The simulated fracture geometry under various fracturing fluid viscosity: (a) 3mPa·s; (b) 5mPa·s; (c) 30mPa·s; and (d) 100mPa·s.

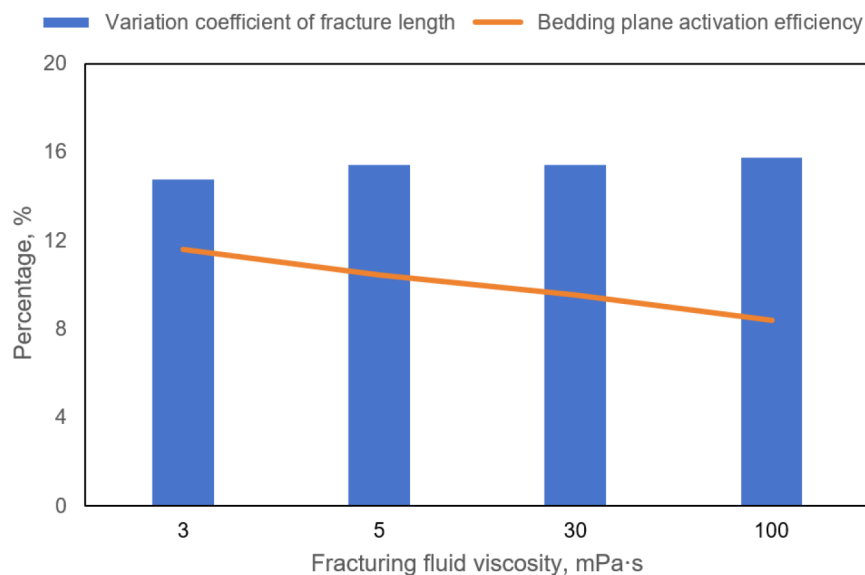


Figure 15—Quantitative evaluation of multiple fracture propagation under various fracturing fluid viscosity: (a) 3mPa·s; (b) 5mPa·s; (c) 30mPa·s; and (d) 100mPa·s.

Case study

The Jurassic Lianggaoshan Formation in the eastern Sichuan Basin exhibits multiple shale intervals, characterized as a normal to abnormally pressured reservoir with high thermal maturity (1.2-1.8%). The shale thickness gradually thins northward from 11 to 27m, composed of black/dark gray shales interbedded with siltstones, where laser scanning shows good oil potential with light oil. These organic-rich shales (avg. TOC 1.8%, S1 3.69mg/g, gas content 1.3cm³/g) display well-developed bedding planes but undeveloped natural fractures, with petrophysical properties of 3.7% porosity and 0.006-0.288mD permeability. Geomechanically, the reservoir contains 44.6% brittle minerals and 43.67% clays on average, exhibiting moderate stiffness (35-44GPa Young's modulus, 0.164 Poisson's ratio) under a strike-slip stress regime ($\sigma_H=65.2\text{MPa}$, $\sigma_V=63.2\text{MPa}$, $\sigma_h=55.1\text{MPa}$).

Well L1 is a horizontal well targeting the Lianggaoshan Formation at a vertical depth of 2638m, employing multi-cluster staged fracturing with intensive 6m cluster space and high injection rates exceeding 18m³/min to maximize SRV. The stimulation utilized a composite temporary plugging method combining knotted fiber diverter and plugging balls to enhance bedding activation and fracture complexity, along with a multimodal proppant scheme (70/140 and 40/70 mesh quartz sand, and 30/50 mesh ceramic proppant) for effective multi-scale fracture filling. The fracturing operation primarily utilizes variable-viscosity slick-water, predominantly low-viscosity formulations, which effectively activates bedding planes and promotes the formation of complex fracture networks.

The well successfully completed 27 fracturing stages with stable treatment pressure curves, showing breakdown pressures of 75.4-89.3MPa and shut-in pressures ranging between 44.45-48.2MPa. The diversion operations achieved clear pressure increments, demonstrating significant net pressure increases. Microseismic monitoring revealed an average fracture length of 299m (32% longer than adjacent wells) with an interpreted height of 24 m, creating an effective SRV that thoroughly covered the sweet spot. Post-fracturing pressure decline analysis indicated a fracture complexity index predominantly exceeding 1.5, suggesting the formation of an effective fracture network comprising both main hydraulic fractures and activated bedding planes. High-frequency pressure monitoring confirmed excellent cluster efficiency (80-100% per stage) with an average conductivity of 12 D·cm, demonstrating successful multi-scale fracture propping, which is beneficial for oil flow. The well began producing oil immediately upon opening and subsequently achieved high production rates during testing.

Conclusion

In this paper, a 3D discrete lattice method is applied to model multi-cluster fracturing in horizontal wells. The effects of geological and engineering parameters on multiple fracture propagation combining the effect of stress interference and bedding planes are studied. The bedding plane activation efficiency and variation coefficient of fracture length are applied to quantitatively evaluate the stimulating effect. The main conclusions are as follows:

- Both experiments and numerical simulation validate the 3D discrete lattice method. The simulation results show that a fence-like fracture network consisting of main fractures and partially activated bedding planes is generated. The geometry of each fracture within a stage is obviously different. The existence of bedding planes will increase the fluid filtration, resulting in reduction of fracture geometry.
- Under the combined effect of bedding planes and stress interference, the bedding planes are more easily activated by hydraulic fracture under the condition of low bedding cohesion, low vertical-horizontal stress difference, small cluster space, high injection rate, and low fracturing fluid viscosity. However, those factors have smaller impacts on the fracture uniformity.

- The research findings of this study have guided the design of fracturing parameters for shale oil horizontal well in the eastern Sichuan Basin, China. The field results demonstrate that the most clusters in this well effectively initiated and propagated, forming a complex fracture network through bedding plane activation, ultimately achieving good oil production.

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